

Formulation

Modeling

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Optimization: principles and algorithms



Modeling

*Translating a concrete problem into a mathematical model is difficult
More of an art, but requires rigor and systematism
This video introduces the process and illustrates it on a small example.*

Motivation

Mathematical model

- ▶ Mathematical representation
- ▶ of a phenomenon,
- ▶ done in order to better study it.

Model for optimization

Decision variables

$$x \in \mathbb{R}^n$$

Objective function

$$f(x) \in \mathbb{R}$$

Constraints

$$x \in X \subseteq \mathbb{R}^n$$

Example: Château Laup-Himum



Problem description

Rosé and red wine using local grapes.

- ▶ Grape: max. 1 ton of Pinot. Price: 3€/kilo.
- ▶ Vinification as rosé: cost 2€ per kilo of grape.
- ▶ Vinification as red (Pinot Noir) : cost 3.50€ per kilo of grape.
- ▶ Price rosé: 15€/ℓ. Rebate: 2€ per 100ℓ.
- ▶ Price red: 23€/ℓ. Rebate: 1€ per 100ℓ.

Château Laup-Himum: decision variables

Liters of rosé to produce per year

x_1

Liters of red to produce per year

x_2

Kilos of grapes to buy

x_3



Château Laup-Himum: objective function



Revenue per liter of rosé

$$15 - \frac{2}{100}x_1$$

Revenue per liter of red

$$23 - \frac{1}{100}x_2$$

Total revenues

$$x_1 \left(15 - \frac{2}{100}x_1 \right) + x_2 \left(23 - \frac{1}{100}x_2 \right)$$

Château Laup-Himum: objective function

Cost for producing rosé

$$2x_1$$

Cost for producing red

$$3.5x_2$$

Cost for buying grapes

$$3x_3$$



Château Laup-Himum: objective function

Total benefits

$$x_1 \left(15 - \frac{2}{100}x_1 \right) + x_2 \left(23 - \frac{1}{100}x_2 \right) - (2x_1 + 3.5x_2 + 3x_3)$$

Château Laup-Himum: constraints



Maximum quantity of grapes

$$x_3 \leq 1000$$

1 kilo of grapes produces 1 liter of wine

$$x_1 + x_2 \leq x_3$$

Non negativity

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

Mathematical model

$$\max_{x \in \mathbb{R}^3} f(x) = x_1 \left(15 - \frac{2}{100}x_1 \right) + x_2 \left(23 - \frac{1}{100}x_2 \right) - (2x_1 + 3.5x_2 + 3x_3)$$

subject to

$$x_1 + x_2 \leq x_3$$

$$x_3 \leq 1000$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

$$x_3 \geq 0$$

Summary

Decision variables, Objective function, Constraints

The set of constraints are often characterized by a set of equations and inequations.

Some constraints like non negativity are easy to forget.